

Progressive Rollout for GCP HCP

Exploring gitops-promoter as a candidate solution

GCP-539 — Tooling spike · May 2026

The Problem (GCP-537)

As we scale to multiple regions, we need to propagate changes safely across environments:

- **No automated promotion pipeline** — changes land directly per environment
- **No enforced ordering** — stage can drift behind or ahead of production
- **No pluggable gating** — CI is branch-level, not change-level
- **ArgoCD runs independently on each cluster** — no cross-cluster API access
- **GitOps constraint** — all promotion must work via Git and external APIs only

| *"Before committing to an implementation, we need to study the landscape of available tools" — GCP-539*

What We Need (GCP-537 Requirements)

- **Environment progression:** integration → stage → production
- **Sectors within environments:** canary sector first, then broader rollout
- **Pluggable gates:** CI checks, health checks, soak timers, manual approval
- **Both promotion models:** individual component bumps and full bundle promotion
- **Observability:** what version is deployed where, promotion history, current readiness
- **Git as source of truth:** all promotion actions result in git commits (auditable, revertable)
- **No cross-cluster API access:** verification via Git and external APIs only

gitops-promoter: What It Is

A Kubernetes controller by **argoproj-labs** that manages promotion via branches and PRs.

- **Source of truth:** a DRY `main` branch; a hydrator renders it per environment
- **Proposed branches** (`env/X-next`): rendered output waiting to be promoted
- **Active branches** (`env/X`): what is currently deployed
- **PRs:** gitops-promoter opens them, gates them with commit statuses, auto-merges (or waits for manual merge)
- **PromotionStrategy CRD:** one resource declares the entire pipeline

Promotion is **declarative** and **auditable** — every promotion is a merged PR with a full CI trail in GitHub.

How It Works

- `main` (DRY source) → **hydrate** → 5 proposed branches (`env/X-next`) pushed simultaneously
- gitops-promoter opens a **PR** for each proposed branch → active branch
- **Proposed** commit statuses (e.g. `ci-check`) gate whether the PR can merge
- Once merged, **active** commit statuses (e.g. `deploy` , `timer`) gate the *next* environment's PR

```
main → hydrate → int/main-next → PR → env/integration/main
                                     | ci-check ✓  deploy ✓
                                     ↓
      → stage/main-next → PR → env/stage/main
                                     | ci-check ✓  deploy ✓  🕒 2m
                                     ↓
      → prod-1-next      → PR → env/production/prod-1 (MANUAL)
                                     | ci-check ✓  deploy ✓
                                     ↓
      → prod-2-next      → PR → env/production/prod-2
                                     | ci-check ✓  deploy ✓
                                     ↓
      → prod-3-next      → PR → env/production/prod-3
```

Demo Pipeline

This POC mirrors the gcp-hcp-infra repo structure across **5 sectors**:

Sector	Regions	Merge
integration/main	us-central1	auto
stage/main	us-central1	auto + 2m soak
production/prod-1	us-central1	MANUAL
production/prod-2	us-central1 · europe-west1	auto
production/prod-3	us-central1	auto

What's promoted each cycle:

- 15 ArgoCD Applications (3 management-cluster + 12 region apps), built with kustomize, packaged as Helm charts
- Terraform configs (platform module + per-sector/region tfvars)
- Per-environment overlays via layered kustomize patches

Gating: Commit Statuses

Status	Type	Set by	What it gates
ci-check	WebRequestCommitStatus	GitHub Actions (YAML lint, terraform validate)	PR merge (proposed)
deploy	WebRequestCommitStatus	GitHub Actions (fake deploy on active branch)	Next sector's PR merge
timer	TimedCommitStatus	gitops-promoter (2-min soak on stage)	prod-1 PR merge

All commit statuses are **standard GitHub commit statuses** — visible on every PR, queriable via API.

WebRequestCommitStatus polls **any HTTP endpoint** — the real implementation would call E2E test results, health check APIs, deployment verification endpoints.

The PromotionStrategy CRD

```
apiVersion: promoter.argoproj.io/v1alpha1
kind: PromotionStrategy
metadata:
  name: gcp-hcp-demo
spec:
  gitRepositoryRef:
    name: gcp-hcp-gitops-promoter-demo
  activeCommitStatuses:      # gates applied to ALL environments
    - key: ci-check
    - key: deploy
  environments:
    - branch: environment/integration/main
    - branch: environment/stage/main
      activeCommitStatuses:
        - key: timer          # 2-min soak before prod-1 unlocks
    - branch: environment/production/prod-1
      autoMerge: false        # requires human approval
    - branch: environment/production/prod-2
    - branch: environment/production/prod-3
```

The controller handles the rest: opens PRs, checks statuses, merges, cascades.

Live Demo

Push a version bump — watch the cascade

Terminal panes:

-

```
watch -n5 promoter/status.sh
```

-

```
watch kubectl -n promoter-system get
```

```
promotionstrategy,commitstatuses,timedcommitstatuses,webrequestcommitstatus,pullrequest
```

Browser: promoter dashboard · GitHub Actions

What We Just Saw

1. Version bump pushed to `main`
2. Hydration workflow built all 5 proposed branches **simultaneously**
3. Integration PR auto-merged once `ci-check` passed
4. Stage PR merged, then **2-minute soak timer** ran before prod-1 unlocked
5. Prod-1 PR required **manual approval** — enforced by the controller
6. Prod-2 and prod-3 **auto-cascaded** after prod-1 merged
7. All 5 sectors now on the same version — every step is a merged PR in Git

Coverage vs GCP-537 Requirements

Requirement	Covered?
Environment progression (int → stage → prod)	✓
Sectors with ordered rollout	✓
Pluggable gates (CI, soak timer, HTTP endpoint)	✓
Manual approval gate	✓
Git as source of truth, all via git commits	✓
Component-level promotion	✓ (this demo)
Bundle / platform-level promotion	❑ (design needed)
Freeze / halt	❑ (not in controller today)
Fast track path	❑ (not in controller today)
Full observability dashboard	❑ (basic dashboard only)

Next: decision document comparing gitops-promoter with other candidates (Kargo, Telefonistka, ...)

→ github.com/patjlm/gcp-hcp-gitops-promoter-demo

→ argoproj-labs/gitops-promoter